SYSDESIGN REFERENCE

Design a Payment System

Requirements-to-architecture reference walkthrough.

Move money through an auditable state machine where retries never duplicate financial effects.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / PAYMENT-SYSTEM

SYSDESIGN

Design a Payment System

CORE IDEA

Move money through an auditable state machine where retries never duplicate financial effects.

Reference architecture decision chain

01	02	03	04
Ledger	Authorization/Capture	Idempotency	Reconciliation

Decisions to defend

+ Ledger
+ authorization/capture
+ idempotency
+ reconciliation
+ webhooks
+ and compliance boundaries.

Failure modes to test

+ Double charge
+ lost webhook
+ ledger mismatch
+ partial refund
+ and provider outage.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Design a Payment System

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS

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